Osptek Display

TFT LCD SPECIFICATION

Model No:

YDP1010BT003-V1

Revision Record 修改记录

Rev No 版本号	Date 时间	Description 内 容
V.0	2025-03-25	First release

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1.General Specifications 基本规格

No.	Item 项目	Specification 规格	Unit 单位	Remark
1	LCD Size 液晶面板尺寸	10.1”	inch	-
2	Panel Type 面板类型	IPS_incell	-	-
3	Resolution 分辨率	800(H)xRGBx1280(V)	Pixel	-
4	Display Mode 显示模式	Normally Black	-	-
5	Number of Colors 颜色数量	16.7M	-	-
6	Viewing Direction 使用视角	ALL	-	Note1
7	NTSC 色彩饱和度	60%	-	Typ.
8	Contrast Ratio 对比度	1500	-	
9	Luminance 亮度	350	cd/m2	Typ.
10	Module Size 模组尺寸	160.00(H)x250.00(V)x3.64(D)	mm	Note1
11	Panel Active Area 可视区域	135.36(H)X216.58(V)	mm	Note1
12	Pixel Pitch 像素尺寸	169.2(H)x169.2(V)	um	
13	Pixel Arrangement 像素排列	RGB-stripe		-
14	Driver IC 驱动芯片	JD9366	-	-
15	Light Source 背光源	32 LED light	-	-
16	Interface 接口方式	MIPI	-	-
17	Operating Temperature 工作温度	-10~+50	℃	-
18	Storage Temperature 存储温度	-20~+60	℃	-

Note 1: Please refer to the mechanical drawing ;

注 1：请参照模组图；

2.Pin Assignments 接口定义

Pin No. Pin 序号	Symbol 符号	Function 功能描述
1	GND	Ground 接地
2	GND	Ground 接地
3	AVDD	Digit Power supply for LCM
4	AVDD	Digit Power supply for LCM
5	NC	
6	AVEE	Digit Power supply for LCM
7	AVEE	Digit Power supply for LCM
8	GND	Ground 接地
9	GND	Ground 接地
10	D3N-	MIPI DSI data lane3-
11	D3P+	MIPI DSI data lane3+
12	GND	Ground 接地
13	D0N-	MIPI DSI data lane0-
14	D0P+	MIPI DSI data lane0+
15	GND	Ground 接地
16	CLKN-	MIPI DSI clock lane-
17	CLKP+	MIPI DSI clock lane+
18	GND	Ground 接地
19	D1N-	MIPI DSI data lane1-
20	D1P+	MIPI DSI data lane1+
21	GND	Ground 接地
22	D2N-	MIPI DSI data lane2-
23	D2P+	MIPI DSI data lane2+
24	GND	Ground 接地
25	GND	Ground 接地
26	ID0	ID0 PIN
27	ID1	ID1 PIN
28	LED PWM	back-light enable signal
29	TE	Tearing effect output
30	RESET	LCM reset signal

2.Pin Assignments 接口定义

31	VDD1.8	Digit Power supply for LCM
32	VDD1.8	Digit Power supply for LCM
33	GND	Ground 接地
34	TP-SCL	TP High speed interface CLOCK differential signal input pins
35	TP-SDA	TP serial data input/output bi-direction pin
36	TP-INT	TP Interrupt
37	TP-RST	TP Reset pin
38	GND	Ground 接地
39	LEDK1-	Backlight cathode 背光负极输入端
40	LEDK2-	Backlight cathode 背光负极输入端
41	LEDK3-	Backlight cathode 背光负极输入端
42	LEDK4-	Backlight cathode 背光负极输入端
43	NC	
44	LEDA+	Backlight anode 背光正极输入端
45	LEDA+	Backlight anode 背光正极输入端

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3. Electrical Specification 电气特性

3 1 Absolute Maximum Ratings 极限参数

Item 项目	Symbol	Value	Unit	Remark
Analog Power Supply Voltage 模拟供电电压	VDD	-0.3~+3.3	V	-
Logic Input Voltage Range	VIN	-0.3~VDD+0.3	V	-
Logic Output Voltage Range	VO	-0.3~VDD+0.3	V	-

3. Electrical Specification 电气特性

3.2 Typical Operation Conditions 典型工作条件

Item 项目	Symbol	Min.最小	Typ.典型	Max.最大	Unit
Supply Voltage 供电电压	VDD1.8	1.65	1.8	2.8	V
Supply Voltage 供电电压	AVDD	5.0	5.5	6.0	V
Supply Voltage 供电电压	AVEE	-6.0	-5.5	-5.0	V
Input High Voltage 输入高电平	V_{IH}	0.7*VDD	-	VDD	V
Input Low Voltage 输入低电平	V_{IL}	DGND	-	0.3*VDD	V
Output High Voltage 输出高电平	V_{OH}	VDD-0.4	-	VDD	V
Output Low Voltage 输出低电平	V_{OL}	DGND	-	DGND+0.4	V

3.3 Backlight Circuit Characteristics 背光功耗

Item	Symbol	Min.	Typ.	Max.	Unit
LED Current 背光电流	I_B	-	80	-	mA
LED Voltage 背光电压	Vf	21.6	24	25.6	V
Power Consumption 功耗	P_{BL}	-	-	-	mW

3.4 LCD Current Consumption 液晶面板功耗

Item	Symbol	Typ.	Max.	Unit
Full Mode 正常模式	VDD	-	-	mW
测试条件：VDD=1.8V； Interface 驱动类型：行翻转或者列翻转； TN Type=>All Black Pattern. TN型液晶面板=>黑色画面； IPS Type=>All White Pattern. IPS型液晶面板=>白色画面； Temperature：25℃；温度：室温25摄氏度；				
Sleep Mode 休眠模式	VDD	-	-	mW
测试条件：VDD=1.8V； DC/DC converter is enabled. Internal oscillator is started and panel scanning is started. 除IC内部晶振和面板扫描外，其他功能都暂停工作； Temperature：25℃；温度：室温25摄氏度；				

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4.Optical Specification 光学参数

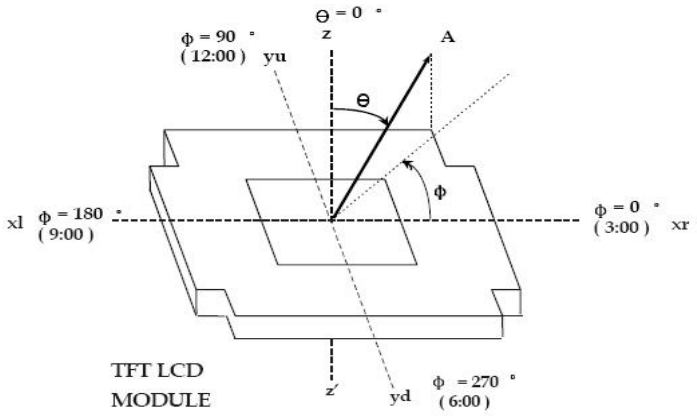
4.1 LCM Optical Characteristics 液晶模组光学特性

Item		Symbol	Condition	Min.	Typ.	Max.	Unit
Viewing Angle Range 视角	Left	θ_L	CR≧10	80	85	-	degree
	Right	θ_R		80	85	-	
	Top	θ_T		80	85	-	
	Bottom	θ_B		80	85	-	
Response Time 响应时间		Ton+Toff	θ=Φ=0°	-	30	35	ms
Contrast Ratio 对比度		CR	θ=Φ=0°	1000	1500	-	-
Luminance 亮度		L	θ=Φ=0°	300	350	-	cd/m^2
Color Chromaticity (CIE1931) 色坐标	White	W_x	Normal	0.24	0.29	0.34	-
		W_y	θ=Φ=0°	0.26	0.31	0.36	
Uniformity 均匀度		U_L	θ=Φ=0°	70	80	-	%

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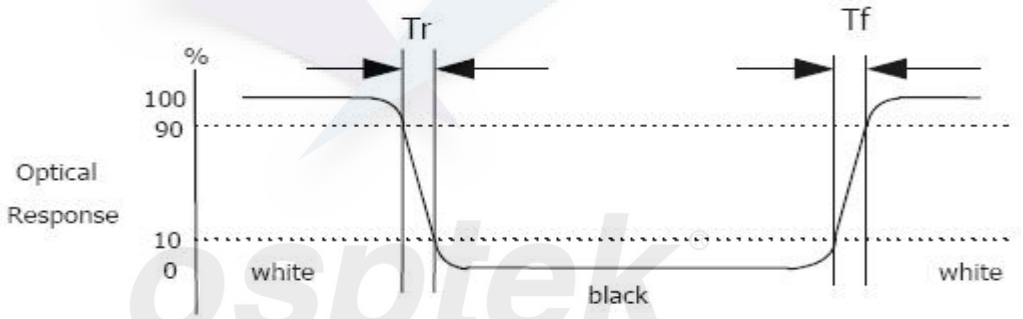
4.2 Measurement system 测量系统

4.2.1 LCM Viewing Angle



Viewing angle is the angle at which the contrast ratio is greater than 10.The angles are determined for the horizontal or x axis and the vertical or y axis with respect to the z axis which is normal to the LCD surface.

4.2.2 Response time



Response time is the time required for the display to transition from white to black (Rising time, Tr) and from black to white (Falling time, Tf) for additional information.

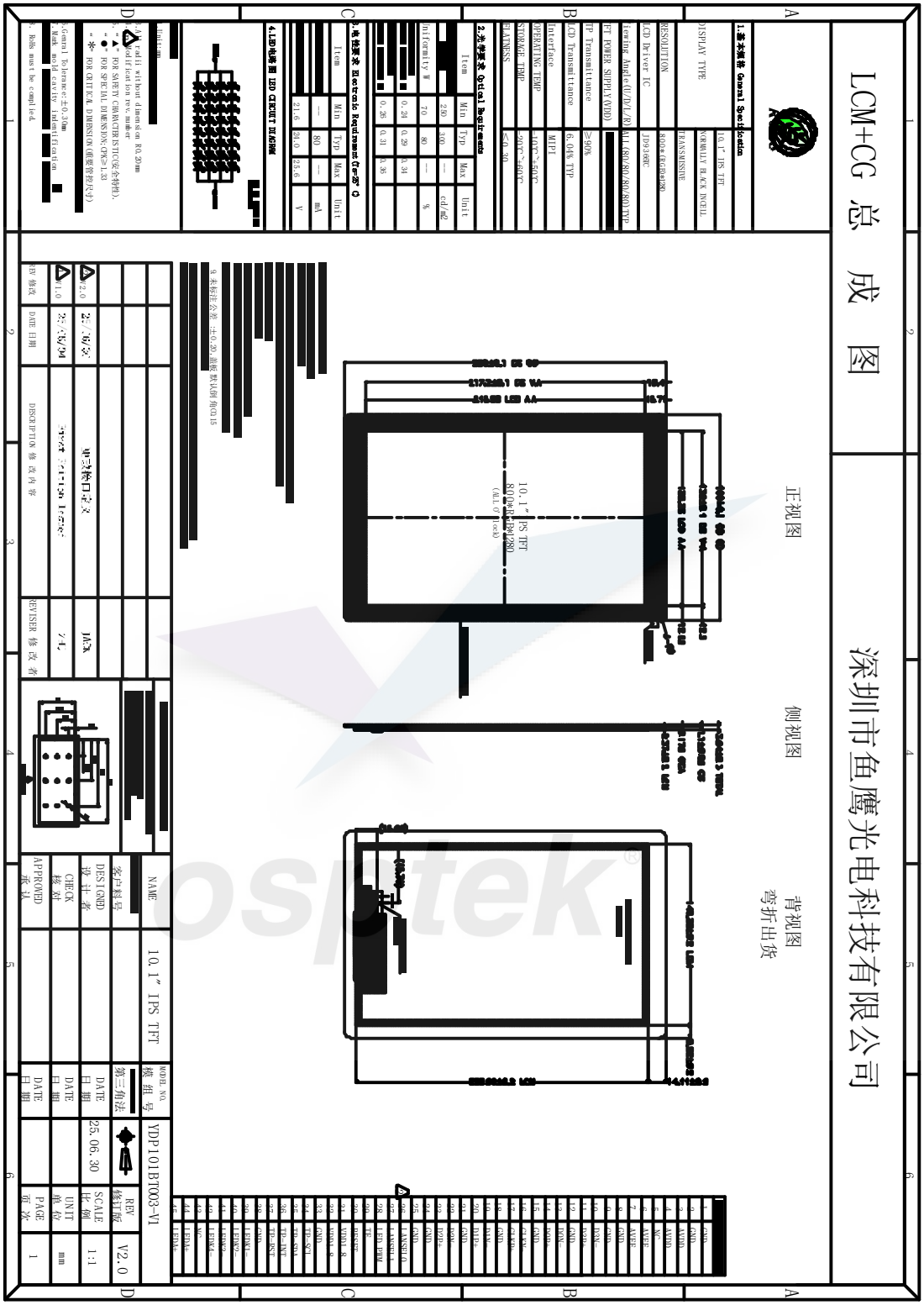
4.2.3 Contrast Ratio (CR)

Contrast Ratio (CR) is defined mathematically as:

$$\text{Contrast Ratio} = \frac{\text{Surface Luminance with all white pixels}}{\text{Surface Luminance with all black pixels}}$$

Surface luminance is the center point across the LCD surface 500mm from the surface with all pixels displaying white.

5. Mechanical Drawing 模组图



6. Reliability Test Items 可靠性测试项目

Test Item 测试项目	Test Condition 测试条件	Test result determinant gist 实验结果判定
High temperature storage 高温存储	60±3℃，72H；	Inspection after 2~4hours storage at room temperature, the sample shall be free from defects: 试验结束后,已测试的LCD样品必须在室内正常温湿度环境下放置2~4个小时以上才能进行功能和外观检查，样品不允许有以下缺陷: 1.Air bubble in the LCD; 模块中有气泡； 2.Non-display; 不显示； 3.Glass crack; 玻璃破碎； 4. The electrical characteristics requirements shall be satisfied. 需要满足模块电气性能。
Low temperature storage 低温存储	-20±3℃，72H；	
High temperature operation 高温运行测试	50±3℃，72H；	
Low temperature operation 低温运行测试	-10±3℃，72H；	
High temperature /humidity 高温高湿	60℃±3℃,90%±3%RH，72H；	
Thermal Shock 冷热冲击	- 20℃/0.5h~+60℃/0.5h for a total 50 cycles；	

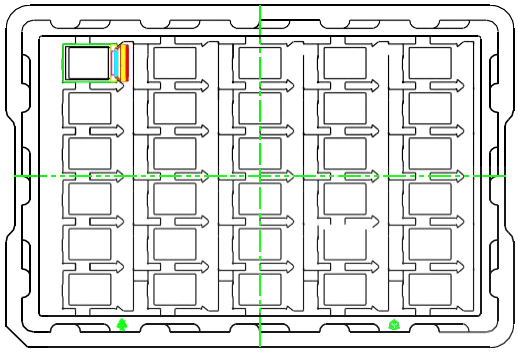
Remark: 注意：

1. The test samples should be applied to only one test item.
   每个被测试的模块只能用于其中的一个测试项目。
2. Sample size for each test item is 2pcs.
   每个测试项目的样品数量为2片。
3. Failure Judgment Criterion: Basic Specification, Electrical Characteristic, Mechanical Characteristic, Optical Characteristic.
   故障判断标准:基本规格,电气特性,机械特性,光电特性。

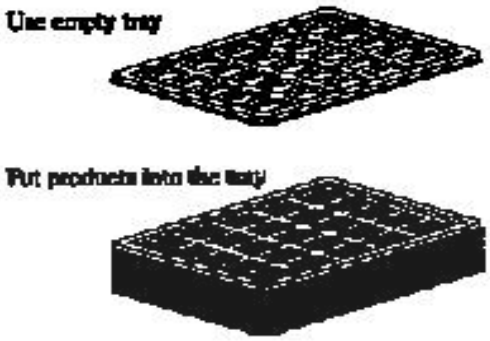
7. Packing and Storage Specification(Reference Only) 包装存储

7.1 Packing Method 包装方法

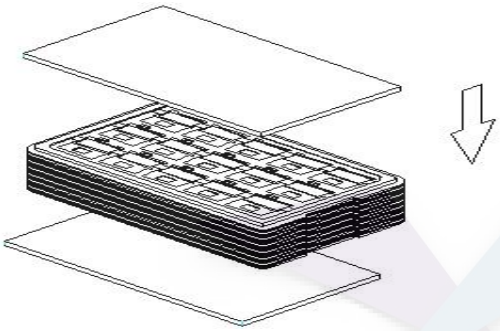
(1)



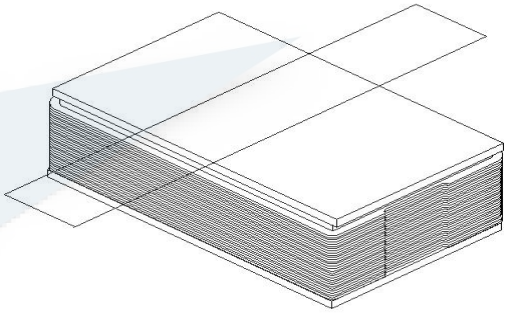
(2)



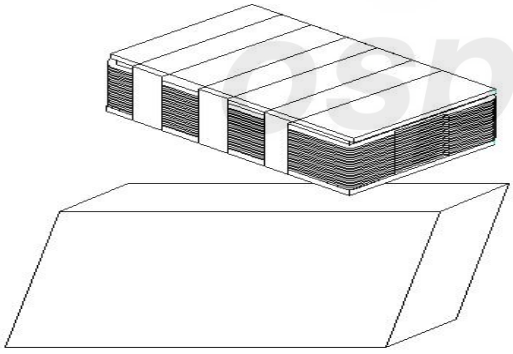
(3)



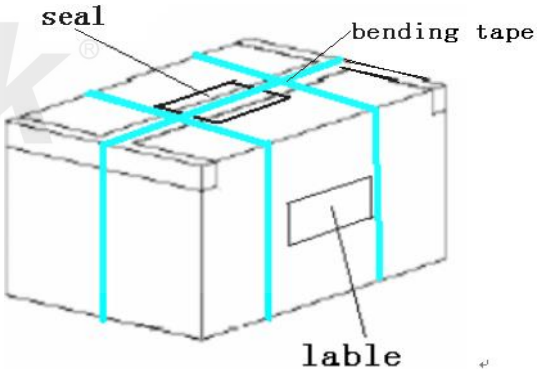
(4)



(5)



(6)



1. Put module into tray cavity. 把模块放进托盘.
2. Tray stacking. 托盘叠装.
3. Put 1 foam under the tray stack and 1 foam above. 在托盘上下放卡板.
4. Fix the cardboard to the tray stack with adhesive tape. 绑胶带.
5. Put the tray stack into carton. 把邦好的托盘放进纸箱.
6. Carton sealing with adhesive tape. 封纸箱.

7.2 Storage Method 存储方法

1. Store in an ambient temperature of 23°C±5°C, and in a relative humidity of 55%±15%. Don't exceed 12 months and expose to sunlight or fluorescent light.
   存储环境温度为 23±5°C，相对湿度为 55%±15%，存储不能超过 12 个月，不要长时间暴晒。
2. Store in a clean environment, free from dust, active gas, and solvent.
   存储在一个干净的环境，不受灰尘，活性气体和溶剂污染。
3. Store in antistatic container.
   存储在防静电环境。

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8. Announcements 注意事项

1. Do not attempt to disassemble or process the LCD module.
   请勿拆卸液晶显示模块。
2. Do not make extra holes on the printed circuit board, modify its shape or change the positions of components to be attached。
   不要在印制电路板上钻额外的孔，修改形状或更改印制线路板上元件的位置。
3. Except for soldering the interface, do not make any alterations or modifications with a soldering iron; Ensure welding temperature at 320 ° C to 350 ° C, the welding time control within the 10 s, welding note don't stay too long in the same place to avoid scald FPC.
   除焊接接口外，不要用烙铁做任何更改；焊接温度保证在 320°C-350°C，焊接时间控制在 10S 以内，焊接时注意不要在同一处停留时间太久以免烫伤 FPC。
4. Other matters in not clear before use, please contact our staff to guide.
   其他事项在不清楚使用之前，请联系我司人员指导进行。

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